AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT132 – M132
Linear Algebra

Final Examination
Spring Semester 2025/2026

Date:6 /June/2026

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Number of Exam Pages: 3 pages (including this cover sheet)	**Time Allowed:** 2 Hours

Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

Q–1: [5×3 marks] Choose the correct answer:

1.1: The determinant of $\begin{bmatrix} 2 & -2 \\ 2 & -2 \end{bmatrix}$

a) -2
b) 2
c) 8
d) 0
e) None of the above

1.2: If $W = Span\left\{\begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ -2 \end{bmatrix}\right\}$, then $\dim(W) =$

a) 0
b) 1
c) 2
d) 3
e) None of the above

1.3: The matrix $\begin{bmatrix} k & 16 \\ 2k & 4k \end{bmatrix}$ is invertible if

a) $k = 0, -8$
b) $k = 0, 8$
c) $k \neq 0, -8$
d) $k \neq 0, 8$
e) None of the above

1.4: If $A = \begin{bmatrix} 3 & 1 & -1 \\ 1 & 4 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -3 & 1 \\ -2 & 0 & 5 \end{bmatrix}$, then

a) $a_{12} + b_{12} = 4$
b) $a_{12} - b_{12} = 4$
c) $a_{21} + b_{21} = 1$
d) $a_{21} - b_{21} = -1$
e) None of the above

1.5: The vectors $\boldsymbol{v_1} = \begin{bmatrix} a \\ 2 \end{bmatrix}, and\ \boldsymbol{v_2} = \begin{bmatrix} 4 \\ 2a \end{bmatrix}$ are linearly dependent if

a) $a = 0, 2$
b) $a = 2, -2$
c) $a \neq 0, 2$
d) $a \neq 2, -2$
e) None of the above

Q–2: [8 marks] Answer the following questions as True or False (2 mark), Justifying your answer (2 marks):

2.1: If A is symmetric and B is an invertible skew-symmetric then $AB^{-1} - B^{-1}A$ is symmetric.

2.2: The linear system $AX = 0$, $A = \begin{bmatrix} 1 & -2 \\ a & 2 \end{bmatrix}$ has infinitely many solutions at $a = -2$.

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Each question is worth 9 marks. Answer the following questions:

Q–3: [5+2+2 marks]

Let $W = \left\{ \begin{bmatrix} 2a-2b \\ a+3b \\ c \end{bmatrix} : c = -a+3b \, , a, b \in \boldsymbol{R} \right\}$ be a subset of $\boldsymbol{R}^3$.

a) Show that W is a subspace of $\boldsymbol{R}^3$.

b) Find a basis for W.

c) Is $\begin{bmatrix} 5 \\ 1 \\ -1 \end{bmatrix} \in W$. Explain.

Q–4: [3+4+2 marks]

Define the mapping $T: \boldsymbol{R^2} \to \boldsymbol{R^3}$ by $T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ 2x+y \\ y \end{bmatrix}$.

a) Show that T is a linear transformation.

b) Find the null of $T, N(T)$ and range of $T, R(T)$.

c) Find $Dim(N(T))$ and $Dim(R(T))$.

Q–5: [9 marks]

The matrix $A = \begin{bmatrix} 3 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix}$ has an eigenvalue $\lambda = 3$. Find the other two eigenvalues then find the corresponding eigenvector of the eigenvalue $\lambda = 3$ and form the diagonal matrix D such that $A = PDP^{-1}$.

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End of questions